Bioinformatic approaches to regulatory genomics and epigenomics

376-1347-00L | week 05

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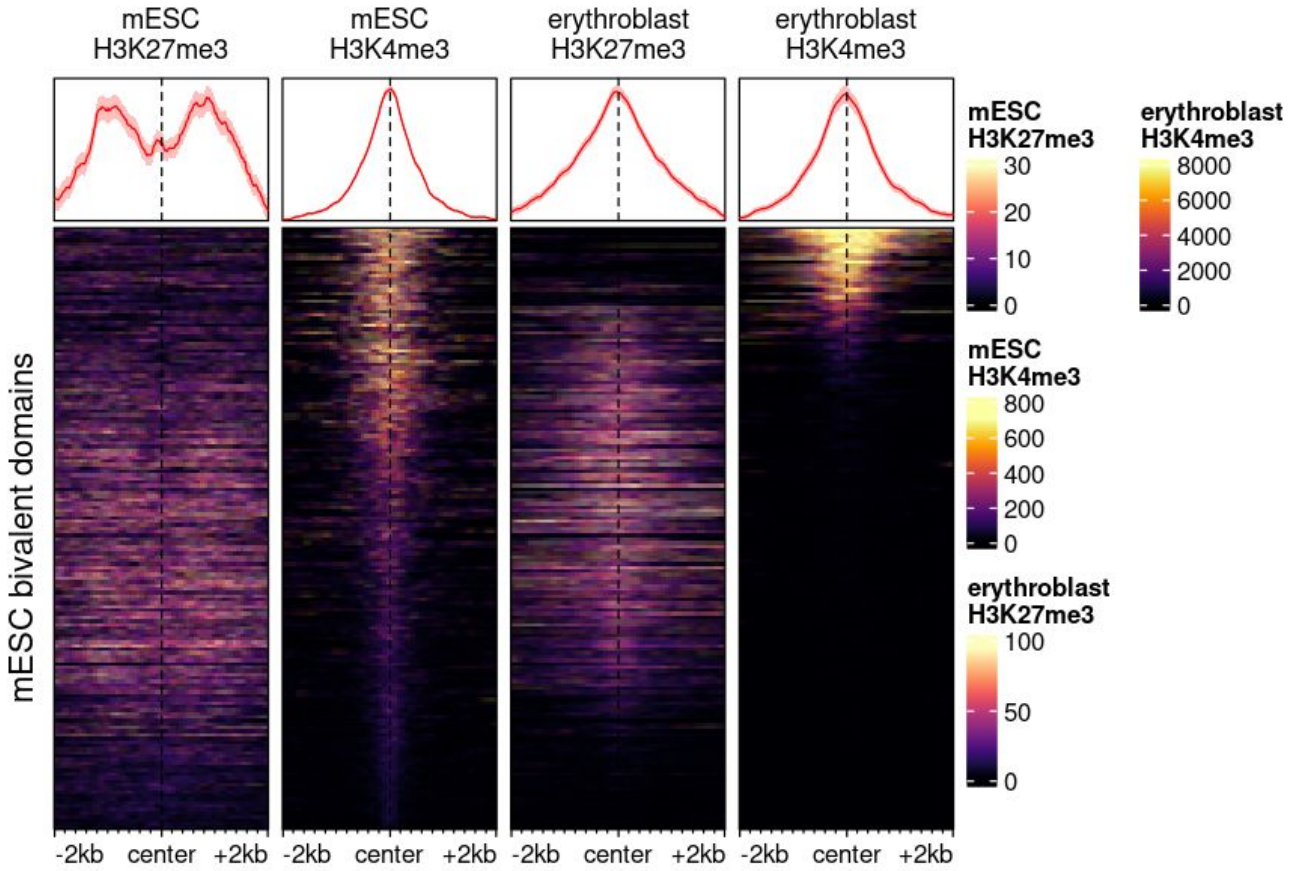
Plan for today

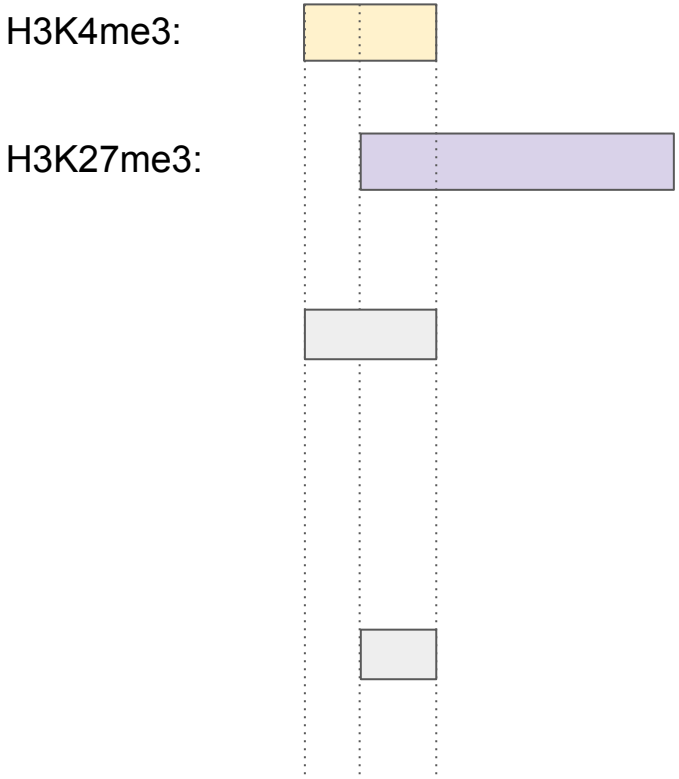
- Debriefing on the assignment
- Statistical testing using (generalized) linear models
- Count-based (discrete) statistics
- Practical:
  - Case study: Yy1 and GDVS
  - Differential binding analysis
- Normalization methods

Embryonic bivalent domains binarize into active and inactive upon differentiation

Classical bivalent TSS
=
H3K4me3
+
H3K27me3



Debriefing: Intersection & overlap
The example of bivalent domains



- **method one (overlapsAny/subsetByOverlaps):**
  find the H3K4me3 peaks that overlap a H3K27me3 domain

- **method two (intersect):**
  find the regions that are covered by both H3K4me3 and H3K27me3

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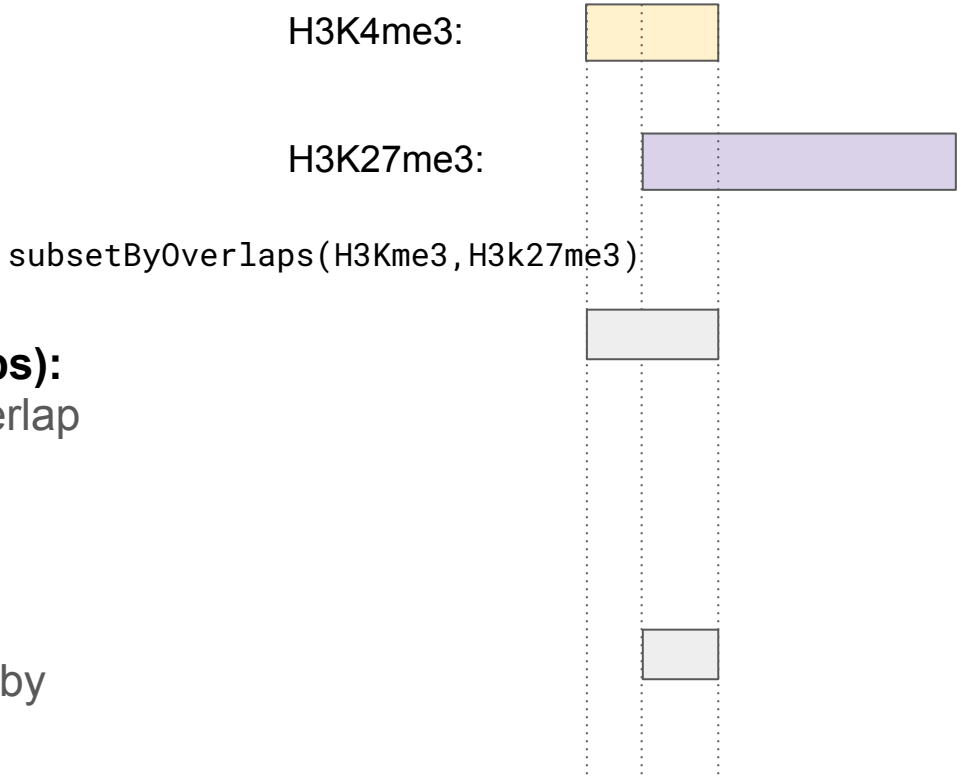

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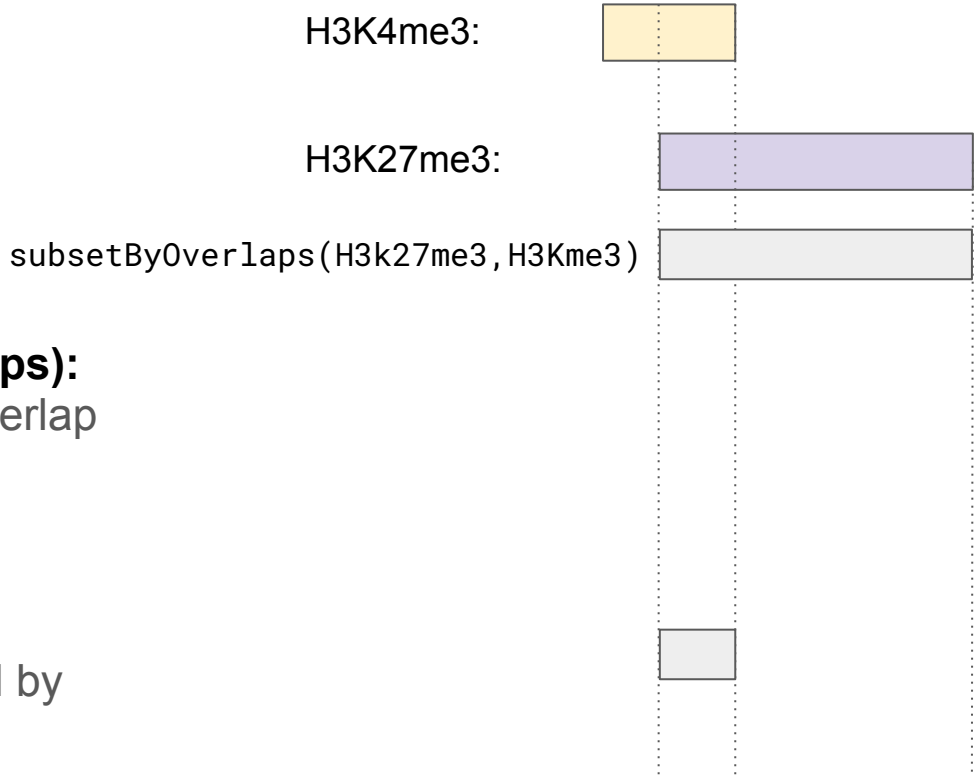

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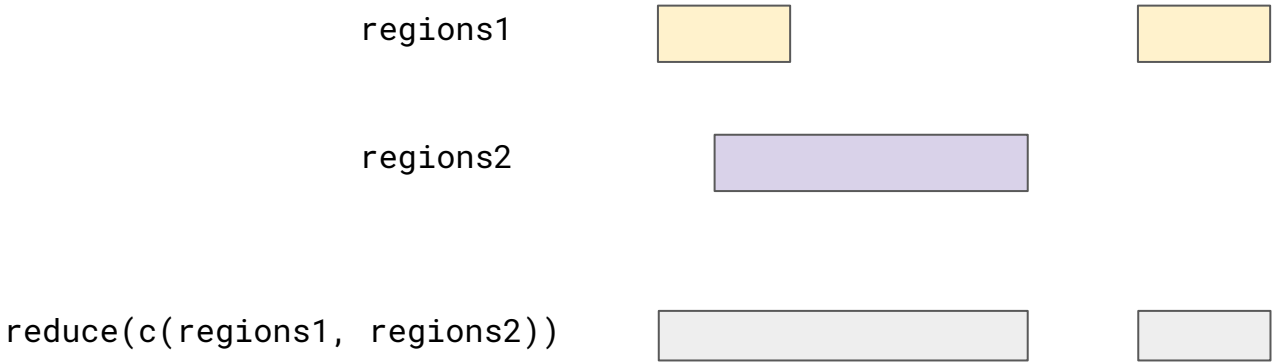
Debriefing on the assignments: Upset Reference

```
regionToUpset(list(regions1, regions2), reference=<...>)
```

Debriefing on the assignments: Upset Reference

When no reference is specified, one is created automatically by merging and *reducing* the regions (unless otherwise specified in the arguments):

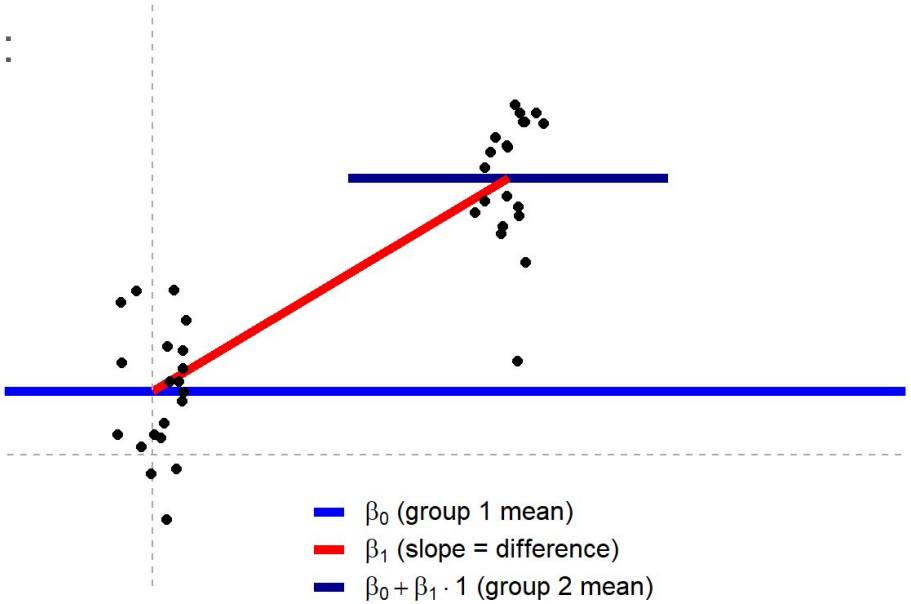


Statistical testing using (generalized) linear models

Working with linear models

Most of the common statistical tests can be formulated as linear regression problems

Consider the *t*-test :



(Taken from [an excellent explanation by Jonas Kristoffer Lindeløv])

$y \sim \beta_0 + x * \beta_1$

Where x=0 for group 1
and x=1 for group 2

Working with linear models - some simple examples

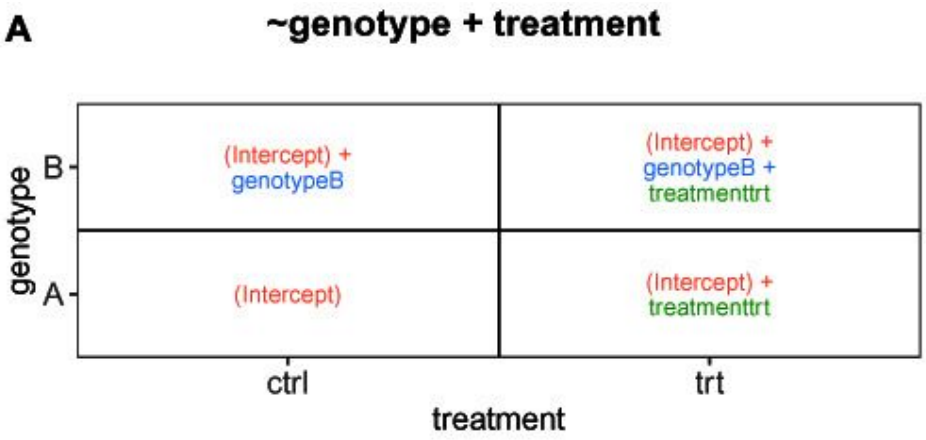
- For a two group comparison:
  - `~group` (equivalent to ~1+`group`)

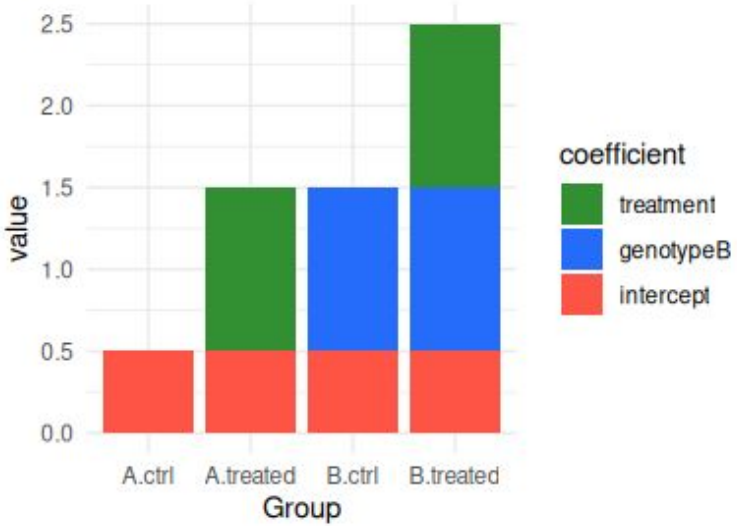
- Comparing between two groups, correcting for the effect of sex:
  - `~sex+group` → then we can still decide to test for the effects of the group by dropping that coefficient

- Finding sex-specific effects
  - `~sex*group` → equivalent to ~1+`sex+group+sex:group`

Working with linear models

Suppose two genotypes (A & B)
and
treatment (trt) vs
control (ctrl)





(Adapted from Soneson et al., *f1000* 2020)

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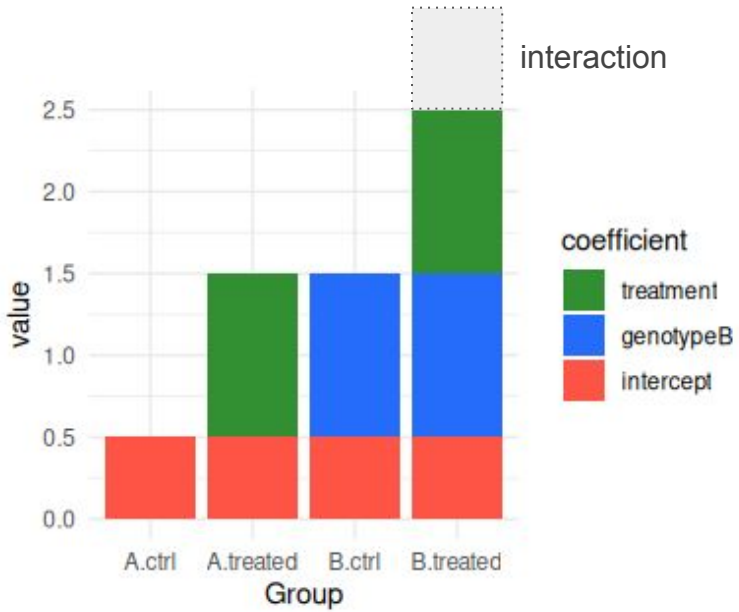


A **~genotype + treatment**

genotype	ctrl	trt
B	(Intercept) + genotypeB	(Intercept) + genotypeB + treatmenttrt
A	(Intercept)	(Intercept) + treatmenttrt

treatment

B **~genotype + treatment + genotype:treatment**

genotype	ctrl	trt
B	(Intercept) + genotypeB	(Intercept) + genotypeB + treatmenttrt + genotypeB:treatmenttrt
A	(Intercept)	(Intercept) + treatmenttrt

treatment

(Adapted from Soneson et al., *f1000* 2020)

2 ways of working with linear models: coefficients & contrasts

Testing coefficients:

```
~ genotype + treatment + genotype:treatment
```

The model matrix has the following coefficients:

```
intercept, genotypeB, treatmentTrt,
genotypeB:treatmentTrt
```

Asking for the effect of the treatment in genotype A amounts to testing coefficient `treatmentTrt`

Testing with contrasts:

Define ‘group’ as the combination of genotype & treatment, i.e. A_ctrl, B_ctrl, A_trt, B_trt

```
~ 0 + group
```

The model matrix has the following coefficients:

```
A_ctrl, B_ctrl, A_trt, B_trt
```

Asking for the effect of the treatment in genotype A amounts to testing the contrast: `A_trt - A_ctrl`

equivalent

For more information on the use of contrasts, see for instance the [limma user guide]

Count-based statistics

The **number of fragments** overlapping a given peak is not a **continuous** variable, but a **discrete** variable: it can only be non-negative integers.

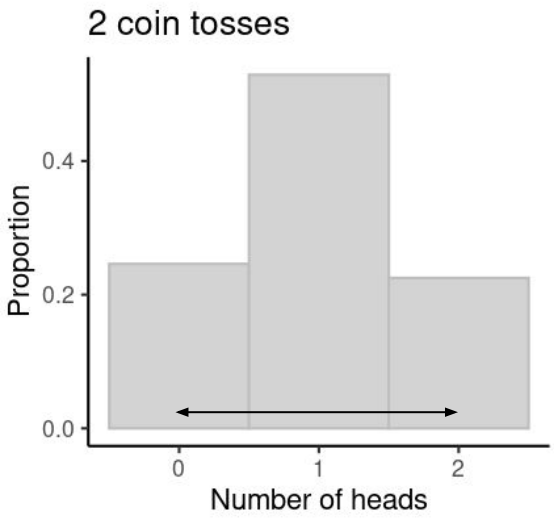
We must consider this information in the statistical analysis of count data, and so we can’t use statistics based on a normal distribution (e.g. t-tests or standard linear models).

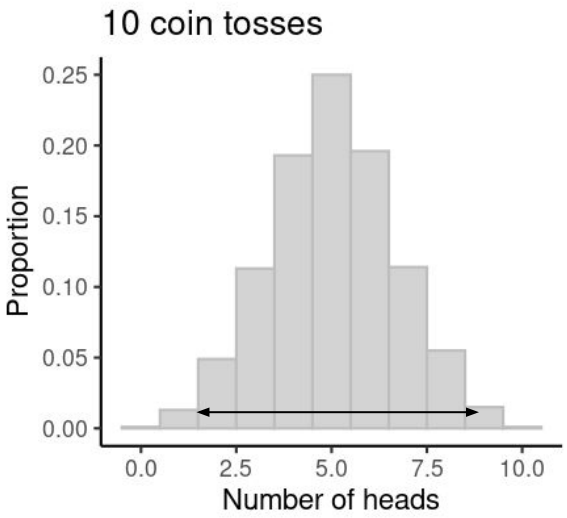
We’ll use **generalized linear models**, which can be applied on different types of data distributions.

Count-based statistics

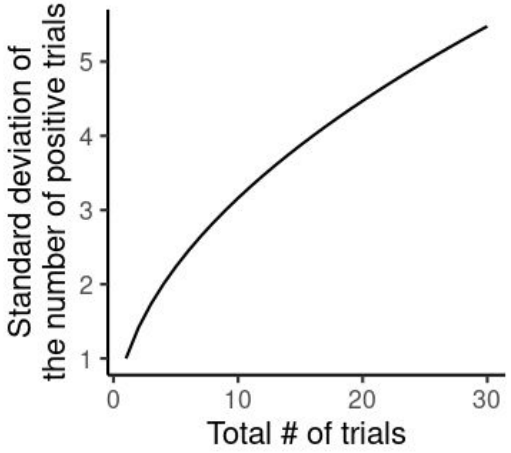
Consider coin tossing, where each toss can result in two outcomes.

Suppose each of us toss a coin X times, and we count the number of ‘heads’.





The results are Poisson-distributed, meaning that the variance is equal to the number of tosses



Count-based statistics

However, counts coming from real phenomena (more complicated than coin tossing) are typically not quite Poisson-distributed: the variance typically increases *more* than would be expected, in other words they show **over-dispersion**.

This is typically because of sources of variation other than the random sampling process, which are typically unknown.

For this reason, count statistics are most often analyzed with the **negative binomial distribution**, which includes an extra dispersion parameter.

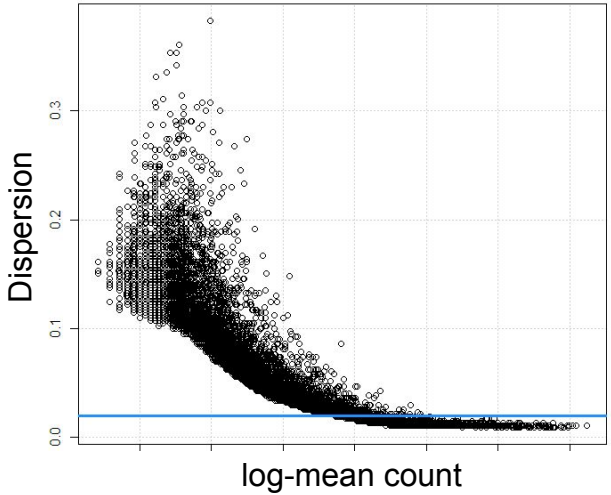
Information-sharing across features

In principle, each feature (e.g. gene or peak) we are analyzing could have a different degree over-dispersion, so that we would have to estimate this parameter separately for each of them.

Typically, we don't have enough samples to do this accurately.

RNAseq analysis packages such as edgeR (and others) therefore rely on a trend between average counts and their overdispersion to moderate those estimates.

We can rely on the same packages for epigenomic data.



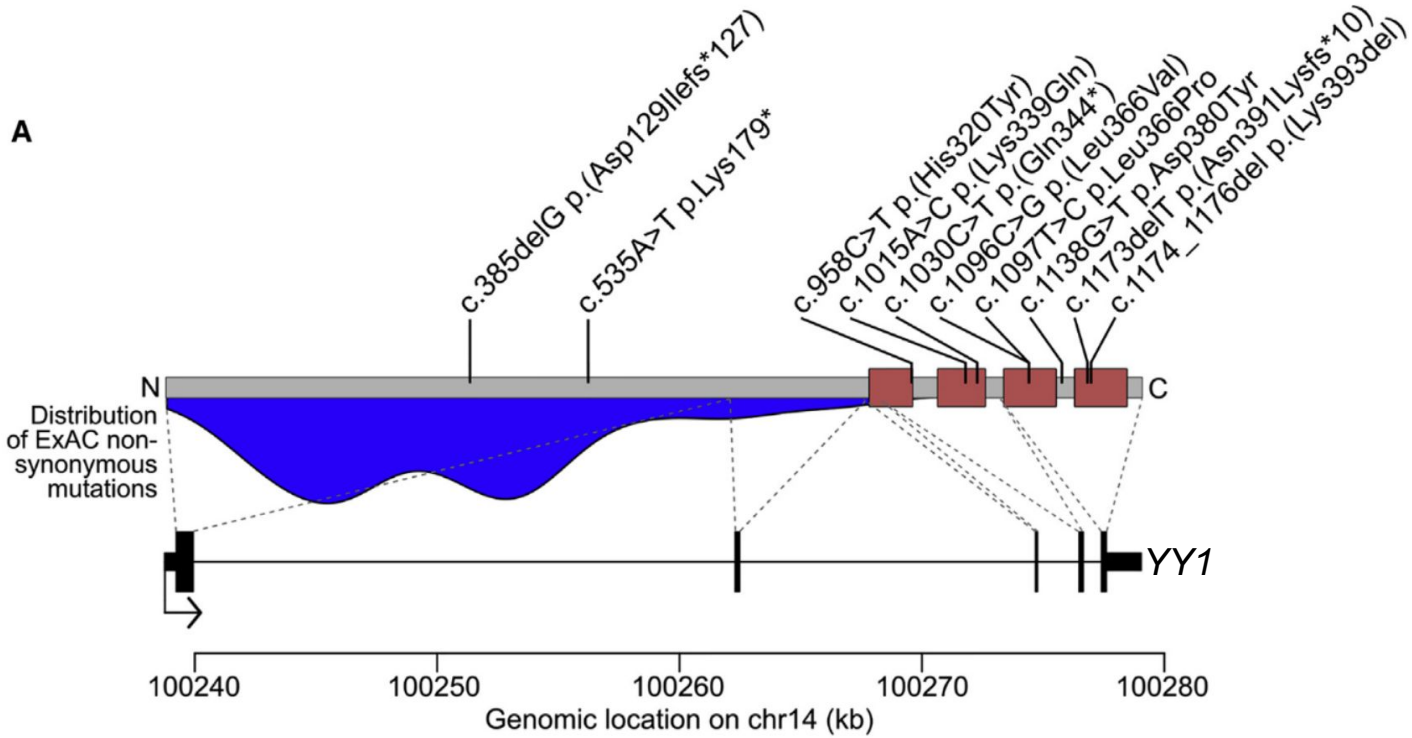
Practical

Our case study today

- Lymphoblastoid cells from patients with Gabriele-de Vries syndrome (and controls)
- OMIM:
  - “Gabriele-de Vries syndrome is an autosomal dominant neurodevelopmental disorder characterized by delayed psychomotor development, variable cognitive impairment, often with behavioral problems, feeding problems, some movement abnormalities, and dysmorphic facial features. Affected individuals may also have a variety of congenital abnormalities.”
- Caused by haploinsufficiency in the *YY1* gene
- Data: YY1 ChIP-seq in mutant and control lymphoblastoid lines

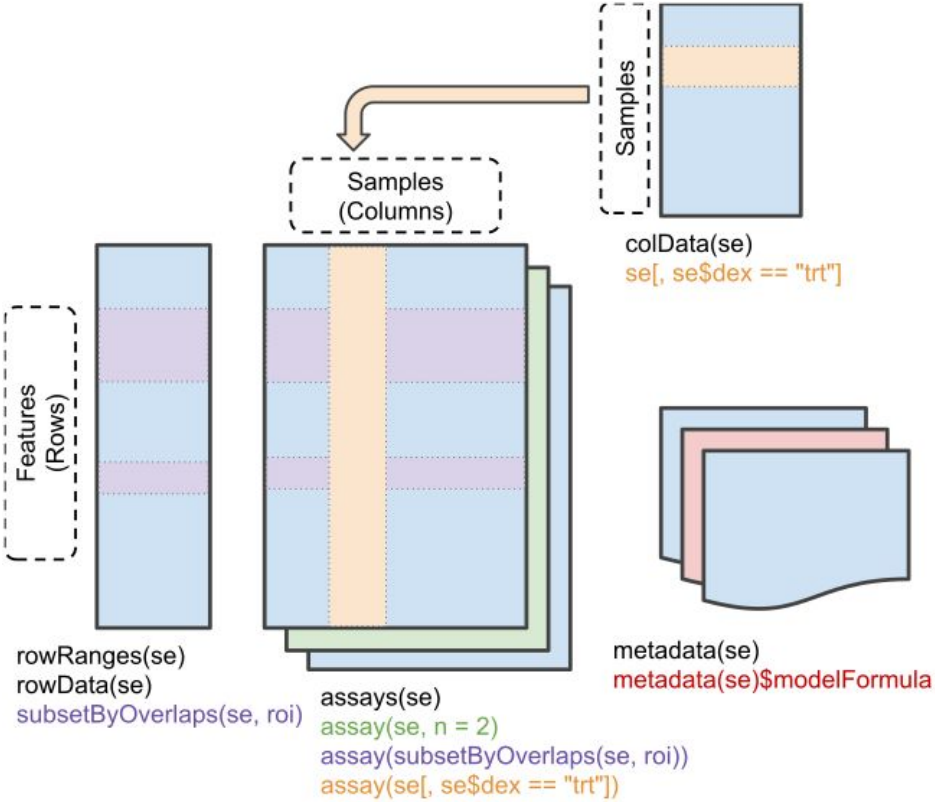
(Gabriele, Vulto-van Silfhout, Germain et al., AJHG 2017)

Our case study today



(Gabriele, Vulto-van Silfhout, Germain et al., AJHG 2017)

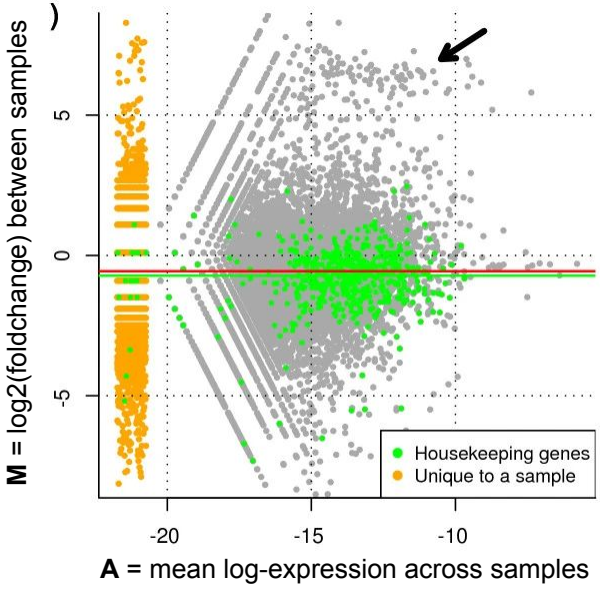
The SummarizedExperiment structure



Documentation

Normalization

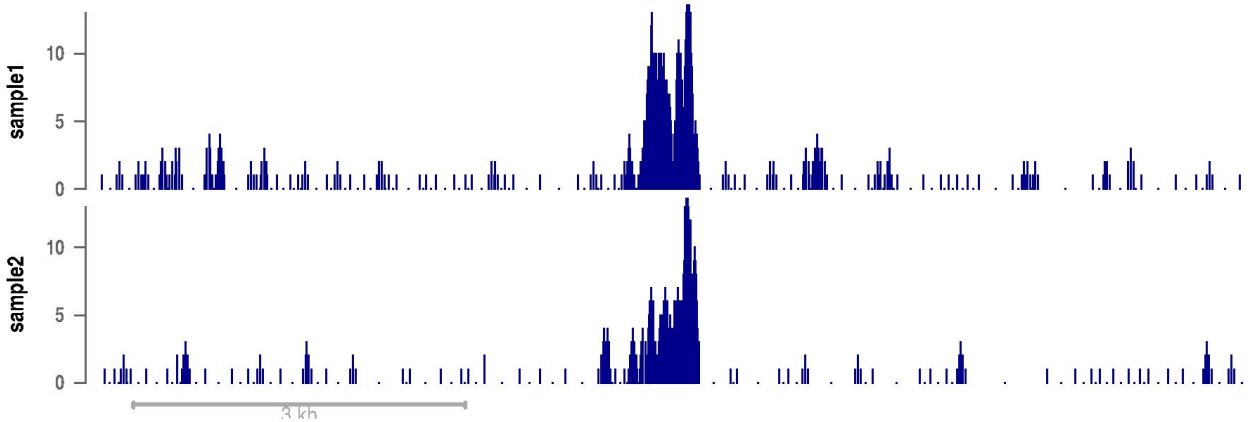
- Standard **TMM normalization** in edgeR assumes that the majority of regions don't change across condition.



(Adapted from Robinson & Oshlack, Genome Biology 2010)

Normalization

- Standard **TMM normalization** in edgeR assumes that the majority of regions (i.e. peaks) don't change across condition.
  - This does not work when there are global increases or decreases (i.e. at most sites) between conditions
- **Background normalization** assumes that the background noise is the same across experiments



Normalization

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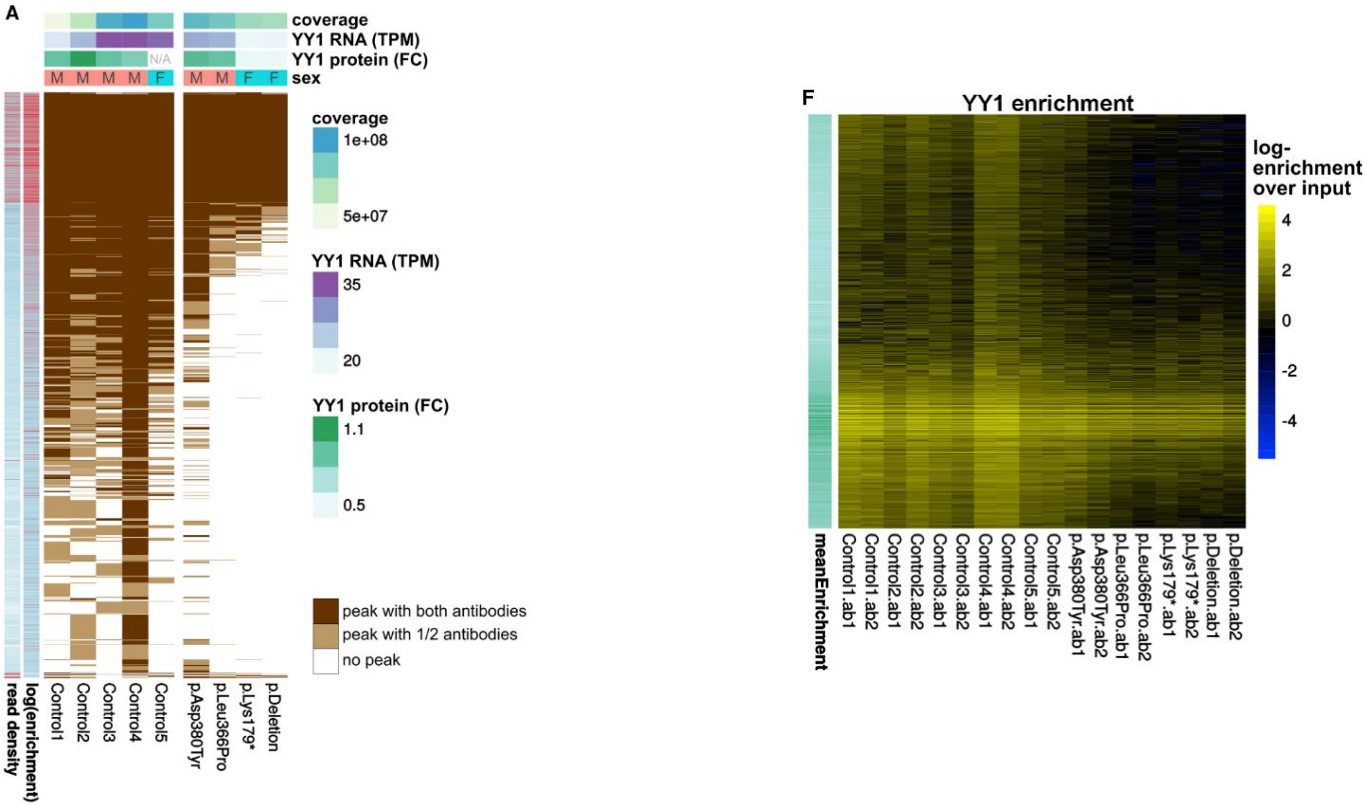
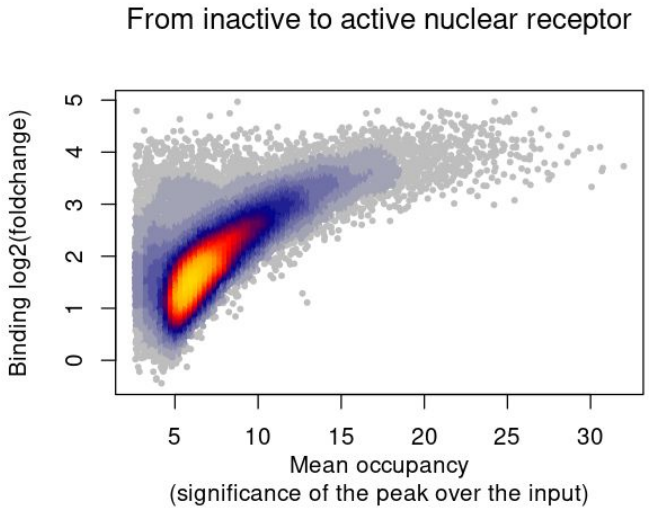
  - This does not work when there are global increases or decreases (i.e. at most sites) between conditions
- **Background normalization** assumes that the background noise is the same across experiments
  - This often works nicely, but can create artifactual differences in highly-enriched regions when the quality of the enrichment differs a lot between experiments
- Common (or top) peak normalization (e.g. **MAnorm**, Shao et al., Genome Biology 2012) assumes that the regions that are significantly enriched in both conditions don't change
- **S3norm** (Xiang et al., NAR 2020) performs both background and common peak normalization

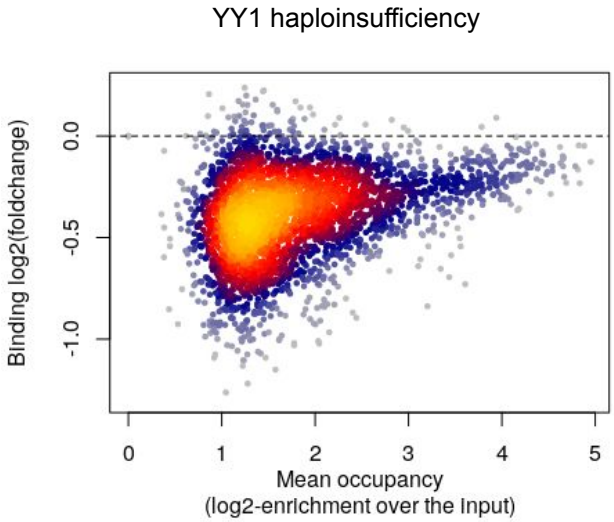
Haploinsufficiency affects especially low-occupancy regions



(Gabriele, Vulto-van Silfhout, Germain et al., AJHG 2017)

When the amount/activity of a TF is changed, not all sites change in binding equally





When activating a TF that wasn't active at all, the TF goes to high-occupancy sites, and these sites tend to show the largest changes in binding

When reducing a TF that's already active, the TF also tends to keep going to the high-occupancy sites, so the low-occupancy sites are those that tend to show the largest change

* co-factors can influence these dynamics!

Further methods for the normalization of technical biases

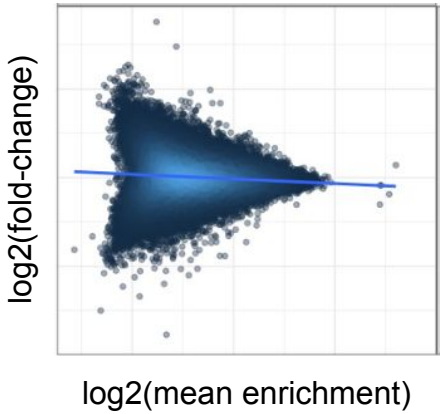
Specifically for ATAC-seq: smooth quantile normalization over GC bins

(Van den Berge et al., *Cell Reports Methods* 2022)

CVnorm

(Unpublished)

loess normalization for enrichment/concentration bias



(see e.g. the *csaw* package – Lun & Smyth, *NAR* 2014)

We'll get back to those when discussing ATAC-seq

Assignment

- Download (a subset of) ATAC-seq peak counts in the hippocampus upon stress (already in SummarizedExperiment format) :
  - https://ethz-ins.org/content/mouse_mm38_hippocampus.peakCounts.SE.rds
  - (the data is from the mouse ensembl GRCm38 genome)
- Using this object, run 2 differential analyses, respectively:
  - comparing stressed (denoted ‘FSS’ – forced swim stress) and control animals
  - comparing male and female animals
- (No need to worry about normalization here, you can just use standard TMM)
- For each analysis, report the top most significant regions, and plot a heatmap of the signal across the samples for those regions